StereoPoseNet: stereo-temporal articulated instrument pose

Current implementation: stereo_pose_unified_c_v1 | 17.945 M parameters | ConvNeXtV2 Nano backbone

Notation: N = 2BT; feature maps = [N, C, H, W]. B: batch, T: time, 2: left/right. Train T=8; streaming T=1.

(a) Shared visual localization and per-camera calibration

Stereo RGB
[N,3,720,1280]
uint8; retain raw image R

Resize + normalize
[N,3,360,640]

ConvNeXtV2 Nano: one shared backbone for both views

Stage 1
2 blocks
[N,80,90,160]

Stage 2
2 blocks
[N,160,45,80]

Stage 3
8 blocks
[N,320,22,40]

Stage 4
2 blocks
[N,640,11,20]

Four-scale feature outputs

Top-down FPN
1x1 lateral projections: C -> 128
Bilinear resize + sum: 4 scales
3x3 Conv 128 -> 128
GroupNorm(8) + GELU
F: [N,128,90,160]

Foreground head: 1x1 Conv 128 -> 1
Mask logits M: [N,1,90,160]

Landmark head: 1x1 Conv 128 -> 5
Heatmaps: [N,5,90,160] for C, H1, H2, J1, J2

Heatmap coordinate decoding
Sharpened sigmoid expectation, power = 4
U: [N,5,2] pixels; confidence: [N,5]

Per-camera K and distortion D
K: [B,T,2,3,3]; D: [B,T,2,5]
Camera vector k: [BT,2,9]

k = [fx/W, fy/H, cx/W, cy/H, k1, k2, p1, p2, k3]
Known per-eye imaging parameters; no stereo baseline input.

Ports below reuse these tensors; the original RGB bypasses the backbone for local crops.

F + M + U

R + M + U

R + M + U

(b) Shaft / body branch

Foreground + five landmark tokens
1 mask-pooled descriptor + C/H1/H2/J1/J2 patches
8x8 sampling -> local encoder -> 4x4 pool
Six ordered tokens: [N,6,128]

Add camera and view embeddings
Camera MLP 9 -> 128 -> 128; reshape [BT,2,6,128]

Bidirectional cross-attention x 2
4 heads, 32 dimensions/head; FFN 128 -> 256 -> 128
Input / output: [BT,2,6,128]

Flatten six tokens; FC 768 -> 128
Reshape: [2B,T,128]

Causal shaft GRU: 128 -> 256
Output [2B,T,256]; h_shaft [1,2B,256]
Shared weights across eyes; separate L/R histories

C token

Rotation heads
Per eye: 256 -> 256 -> 6
6D -> R: [B,T,2,3,3]

Depth heads
[h(256), C token(128)]
384 -> 256 -> 1; softplus

C = Z [xn, yn, 1]
C pixel from U + K,D
Brown inverse -> ray
C: [B,T,2,3]

C_L, R_L | C_R, R_R

(c) Wrist branch

Predicted foreground crop from raw RGB
Detach mask and ROI coordinates; mask RGB before sampling
ROI uses C + hinge + both tips
RGB + mask: [N,4,160,160]

Independent lightweight CNN
4 -> 32 -> 48 -> 64 channels; strides 2,2,1
[N,32,80,80] -> [N,48,40,40] -> [N,64,40,40]
Depthwise + pointwise residual refinement

Spatial descriptor + local metadata
Pool 4x4 -> 1024 -> FC 128
Add metadata MLP 15 -> 128 -> 128
Paired descriptors: [BT,2,128]

Local stereo fusion: concatenate + MLP
[own(128), other(128)] -> 256 -> 128 -> 128
Add own-view descriptor; [BT,2,128]
Reshape to temporal sequences: [2B,T,128]

Causal wrist GRU: 128 -> 256
Output [2B,T,256]; hidden state [1,2B,256]
Own parameters; shared across L/R, separate L/R histories

Concatenate L/R states -> angle head
256 + 256 = 512 per timestamp
FC 512 -> 256 -> 2; sigmoid into physical joint limits
pitch, yaw: [B,T,2]

pitch, yaw

(d) Paired-jaw branch

Predicted foreground crop from raw RGB
Detach mask and ROI coordinates; mask RGB before sampling
ROI uses hinge + both tips
RGB + mask: [N,4,160,160]

Independent lightweight CNN
4 -> 32 -> 48 -> 64 channels; strides 2,2,1
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Local stereo fusion: concatenate + MLP
[own(128), other(128)] -> 256 -> 128 -> 128
Add own-view descriptor; [BT,2,128]
Reshape to temporal sequences: [2B,T,128]

Causal jaw GRU: 128 -> 256
Output [2B,T,256]; hidden state [1,2B,256]
Own parameters; shared across L/R, separate L/R histories

Concatenate L/R states -> angle head
256 + 256 = 512 per timestamp
FC 512 -> 256 -> 1; sigmoid into physical joint limits
jaw half-angle: [B,T,1]

jaw half-angle

(e) Pose output and deterministic articulated geometry

Exported articulated pose: [B,T,17]
C_L(3) + quaternion_L(4) + C_R(3) + quaternion_R(4)
+ wrist pitch(1) + wrist yaw(1) + jaw half-angle(1)
Quaternion order: xyzw; C: mm; joint angles: radians.

Known CAD / USD kinematic chain
Derive d, H1/H2, J1/J2, T and three local XYZ frames
Project each eye with its own K,D -> arrows and panels
Symmetric jaw joints: q1 = -half, q2 = +half

Three independent temporal modules; state pack [3,2B,256] = [shaft, wrist, jaw]. No future frames are used.

R = original RGB; F = FPN features; M = mask logits; U = predicted landmark pixels. Dashed route = reused C token.

Tensor dimensions checked at B=1, T=1; the architecture supports causal streaming with three independent states.

How stereo evidence and temporal history are fused

Operator-level details of the same model. Body attention and local-part MLP fusion are distinct implementations.

N = 2BT. Every branch shares weights across views but keeps the left and right recurrent states separate.

(a) Shaft: bidirectional token attention

Left tokens L
[BT,6,128]

Right tokens R
[BT,6,128]

LayerNorm(128)
Same normalization parameters for L/R

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Q = L

Q = R

Left <- right attention
Q: L; K,V: R
4 heads x 32; sequence length = 6
Residual: L + attention output

Right <- left attention
Q: R; K,V: L
4 heads x 32; sequence length = 6
Residual: R + attention output

Residual feed-forward network
LN128 -> FC128:256 -> GELU
-> FC256:128; add residual

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Two independently parameterized blocks; output [BT,2,6,128]

Six-token pooling -> causal shaft GRU
Flatten 768 -> FC128 -> LN128 -> GELU -> [2B,T,128]
GRU128:256 -> per-eye rotation heads; state + C token -> depth heads

Both attention directions read the original pair inside each block.
Token order: foreground, C, H1, H2, J1, J2. C token is also routed to Z.

(b) Wrist and jaw: two independent local branches

Local left descriptor fL
CNN descriptor(128) + metadata(128)
[BT,128]

Local right descriptor fR
CNN descriptor(128) + metadata(128)
[BT,128]

[fL || fR] : 256
LN256 -> FC256:128 -> GELU
-> FC128:128; add fL
Fused left descriptor: [BT,128]

[fR || fL] : 256
LN256 -> FC256:128 -> GELU
-> FC128:128; add fR
Fused right descriptor: [BT,128]

Left causal GRU state
Input [B,T,128] -> [B,T,256]
hL(t) = GRU(xL(t), hL(t-1))

Right causal GRU state
Input [B,T,128] -> [B,T,256]
hR(t) = GRU(xR(t), hR(t-1))

Same GRU weights within a branch; no sharing between wrist and jaw

Late stereo-state concatenation
[hL(t) || hR(t)] -> [B,T,512]
FC512:256 -> GELU -> FC256:k -> sigmoid joint bounds

Wrist: k = 2
pitch / yaw: [B,T,2]

Paired jaw: k = 1
half-opening: [B,T,1]

(c) Building the six body tokens

F: [N,128,90,160] | M: [N,1,90,160] | U: [N,5,2]
U from sigmoid(heat)^4 weighted coordinate expectation; confidence = max sigmoid(heat)

Whole-instrument token
w = sigmoid(M)^2 / sum
Weighted spatial pooling of F
[N,1,128]

Five landmark-centered feature patches
Radius 24 network pixels; sample 8x8
grid_sample -> [5N,128,8,8]
align_corners=False; border padding

Lightweight local residual encoder
DWConv3x3(128) -> GN8 -> GELU -> PWConv1x1
Add sampled patch; pool to [5N,128,4,4]
Flatten 2048 -> FC128 -> LN128 -> GELU
Local tokens: [N,5,128]

Concatenate + positional / camera embeddings
Whole(1) + local(5) -> [N,6,128]; add region embedding [6,128]
Per-token metadata: 5 -> 128 -> 128; add camera: 9 -> 128 -> 128
Add learned view embedding [2,128]; reshape [BT,2,6,128]

Whole metadata: center(2), spread(2), mean mask(1). Local metadata: uv(2), radius(2), confidence(1).

(d) Native-resolution crop and local CNN

Predicted foreground routing: stop gradients through mask and ROI
sigmoid(M) -> resize 720x1280 -> clamp((p-0.2)/0.8,0,1) -> max-pool3x3
Normalize source RGB; multiply by gated mask BEFORE bilinear ROI sampling

Wrist ROI: C,H,J1,J2 | Jaw ROI: H,J1,J2 | H=(H1+H2)/2
Concatenate foreground RGB + mask; sample [N,4,160,160] for each branch
align_corners=False; zero padding; crop coordinates derived from predicted U

Conv3x3 / stride 2
4 -> 32; GN8 + GELU
[N,32,80,80]

Conv3x3 / stride 2
32 -> 48; GN8 + GELU
[N,48,40,40]

Conv3x3 / stride 1
48 -> 64; GN8 + GELU
[N,64,40,40]

Residual refinement: DWConv3x3(64) -> GN8 -> GELU -> PWConv1x1
Feature remains [N,64,40,40]; adaptive pool4x4 -> [N,1024]
FC1024:128 -> LN128 -> GELU -> local descriptor [N,128]

Add local metadata MLP 15:128:128
K/D(9) + box(4) + visibility(1) + conf(1)

Auxiliary heatmap: Conv1x1 64 -> 3
[N,3,40,40]; training/debug output

Temporal state contract

h_shaft [1,2B,256] + h_wrist [1,2B,256] + h_jaw [1,2B,256] -> [3,2B,256]

Each frame advances all three modules once. Reset on sequence / timestamp / calibration discontinuity; preserve state within a sequence.

Training uses 8-frame windows. Streaming supplies one image pair at a time, with history retained in the three states.

Solid arrows: feature flow. Dashed stereo arrows: other-view input. Auxiliary local heatmaps are skipped by the normal optimized runtime.